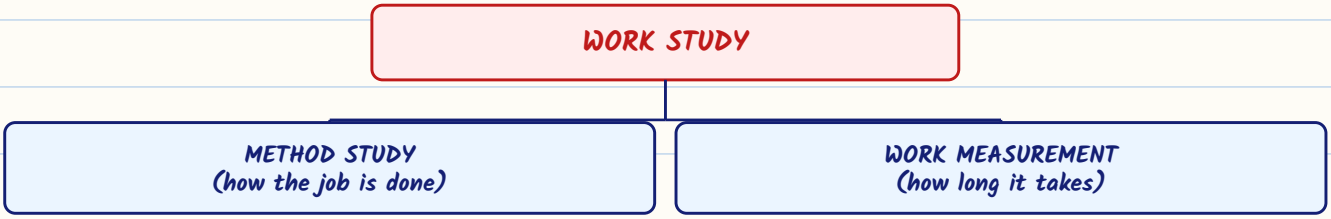
UNIT - II : WORK STUDY

I. Introduction and importance

* WORK STUDY = systematic examination of the methods of carrying out activities so as to improve effective use of resources and to set up standards of performance for the activities carried out (ILO). It is the foundation of productivity improvement and is also called 'Methods Engineering'.
* Two parts: METHOD STUDY (motion study) - finds the best way of doing a job; WORK MEASUREMENT (time study) - finds the time a qualified worker takes to do it by the best method. Method study first, then work measurement.



Importance of work study

* Increases productivity without capital investment; lowers cost; gives standard times for production planning, scheduling and costing; basis for wage incentive plans; eliminates waste and unnecessary movement; better layout and material handling; reduces fatigue, improves safety and working conditions; fair workload and job satisfaction.

Types of production and application of work study

Type	Features	Work study use
Job production	One / few items to customer order, general purpose machines, skilled labour	Less detailed; method study on important jobs
Batch production	Moderate quantity in lots, repeated at intervals	Time standards for each batch operation
Mass (flow) production	Large volume of standard product, special purpose machines	Detailed method and time study; line balancing
Continuous (process) production	Non stop, chemicals, oil, cement, sugar	Process analysis, work sampling

* Work study applies to manufacturing, offices (clerical work), hospitals, banks, maintenance, construction, warehouses, hotels and retail.
* PROCEDURE (SREDIM): Select the job, Record all facts, Examine critically, Develop the best method, Install it, Maintain it.

2. Method Study

* METHOD STUDY = systematic recording and critical examination of existing and proposed ways of doing work, as a means of developing and applying easier and more effective methods and reducing costs.

Steps (SREDIM)

Step	Action
S - Select	Choose the job: high cost, bottleneck, high labour content, repetitive, hazardous, complaints
R - Record	Record all relevant facts by charts, diagrams, films
E - Examine	Critical examination: ask Purpose? Place? Sequence? Person? Means? -> Why? ; Eliminate, Combine, Rearrange, Simplify (ECRS)
D - Develop	Develop practical, economical, effective method
I - Install	Get approval, train workers, put into practice
M - Maintain	Follow up, check that the method is not reverted

Recording techniques – Process charts and symbols



Operation Inspection Transport Delay Storage

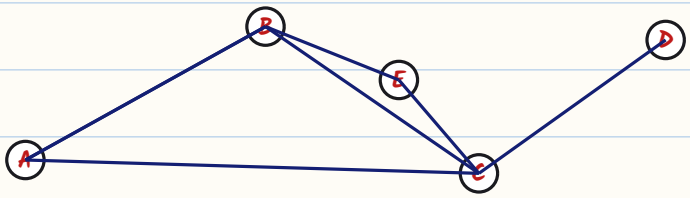
* OUTLINE PROCESS CHART: only operations and inspections (overall picture). FLOW PROCESS CHART: all five symbols - operation, inspection, transport, delay, storage; three types - MAN type (what the worker does), MATERIAL type (what happens to material), EQUIPMENT type. Includes distance moved and time.

Symbol	Meaning	Eg.
Circle	Operation - changes the object (productive)	Turning a shaft
Square	Inspection - check quality or quantity	Gauging diameter
Arrow	Transport - move from one place to other	Carry by trolley
D (big)	Temporary delay - waiting	Waiting in queue
Inverted triangle	Storage - controlled keeping	Raw material store
Circle in square	Combined operation + inspection	Machine and gauge

Only OPERATION is value-adding; transport, delay, storage, inspection are non-value-adding - to be eliminated or reduced by method study.

Flow diagram and string diagram

* FLOW DIAGRAM: scale plan of the shop on which the path of movement of the worker / material is drawn with the process chart symbols. STRING DIAGRAM: a scale model of the layout; a thread is passed round pins at work points to show the movement paths; the length of thread measures the total travel. Shows congestion and backtracking.



String diagram: scale layout of the shop; a thread is wound round pins at work points to trace movement paths of worker / material. Most thread between two points -> place them nearer.

Example: flow process chart (man type) - drilling a component

Step	Description	Symbol	Dist (m)	Time (min)
1	Take casting from bin	Operation	-	0.20
2	Carry to drilling machine	Transport	12	0.40
3	Wait for machine	Delay	-	0.50
4	Clamp and drill holes	Operation	-	1.80
5	Check hole diameter	Inspection	-	0.30
6	Carry to store	Transport	20	0.60
7	Store in tray	Storage	-	-

Summary: operations 2, transport 2 (32 m), inspection 1, delay 1, storage 1. Improvement by ECRS: place machine nearer bin (cut transport), keep a second casting ready (remove delay).

* Critical examination - PRIMARY questions: What is done? Why? Where? Why there? When? Why then? Who does it? Why that person? How? Why that way? -> SECONDARY questions then ask: what else could be done, where else, when else, who else, how else -> choose what SHOULD be done.

Other recording tools

* MULTIPLE ACTIVITY (man-machine) chart: time scale chart showing activities of several workers / machines to find idle time (man-machine ratio, number of machines per operator). SIMO chart (simultaneous motion cycle): left and right hand motions of a worker on a common time scale using therbligs. Travel chart (from-to chart), cyclegraph (photograph of light on moving hand), chronocyclegraph (interrupted light - speed), film analysis (memo-motion, micro-motion).

Motion study and the principles of motion economy

* MOTION STUDY (Gilbreths) = analysis of the body motions used in doing a job to eliminate wasted motion and set up the best motion sequence. Frank Gilbreth identified 17 basic divisions of human motion called THERBLIGS (Gilbreth spelt backwards, with th transposed).

Therbligs

Therblig	Symbol	Class	Therblig	Symbol	Class
Search	Sh	Ineffective	Hold	H	Ineffective
Select	St	Ineffective	Rest for overcoming fatigue	R	Ineffective
Grasp	G	Effective	Unavoidable delay	UD	Ineffective
Transport empty	TE	Effective	Avoidable delay	AD	Ineffective
Transport loaded	TL	Effective	Plan	Pn	Ineffective
Position	P	Effective	Inspect	I	Effective
Assemble	A	Effective	Release load	RL	Effective
Disassemble	DA	Effective	Pre-position	PP	Effective
Use	U	Effective			

* Aim: eliminate ineffective therbligs (search, select, plan, hold, delays, rest) and shorten the effective ones.

Principles of motion economy (Barnes)

* (a) Use of the HUMAN BODY: both hands begin and end motions together; hands not idle together except rest; motions of arms symmetrical and in opposite directions; use the lowest classification of body movement (finger < wrist < forearm < upper arm < shoulder / body); momentum helps; smooth continuous curved motion preferred over zig-zag; rhythm.
* (b) ARRANGEMENT of the WORKPLACE: fixed place for tools and

materials; tools pre-positioned; gravity feed bins and drop delivery; materials and tools in order of sequence; proper lighting, seating, height of work.

* (c) Design of TOOLS and EQUIPMENT: use jigs and fixtures instead of hands for holding; combine two or more tools; foot operated devices; handles designed for maximum contact; levers and wheels within easy reach.

Micro-motion study

* Detailed study of short-cycle repetitive operations by filming the work at constant speed (e.g. 1000 frames per minute / 16-24 fps) along with a MICROCHRONOMETER (clock) in the picture; the film is analysed frame by frame to break the job into therbligs and time each element in WINKS (1 wink = 1/2000 min). Used for highly repetitive, mass production work; costly. Memo-motion: filming at slow speed (1 frame / second) for long cycle jobs.

3. Work Measurement (Time Study)

* WORK MEASUREMENT = application of techniques to establish the time for a QUALIFIED worker to carry out a specified job at a defined level of performance. Uses: production planning, labour cost, incentive plans, manpower and machine requirement, delivery dates, comparing methods.
* Techniques: (1) time study (stopwatch), (2) work sampling (ratio delay), (3) synthesis from standard data, (4) predetermined motion time systems (PMTS), (5) analytical estimating, (6) historical records.

Time study procedure

* 1. Select the job and worker (average qualified worker). 2. Record information on the study sheet. 3. Break the job into ELEMENTS (short, identifiable). 4. Time each element with a stopwatch (cumulative / snap-back / continuous method) for a number of cycles. 5. RATE the worker's performance. 6. Compute the normal time. 7. Add ALLOWANCES to get the standard time. 8. Check and publish.
* Number of observations required: n = (40 sqrt(N sum(x^2) - (sum x)^2) / sum(x))^2 at 95% confidence and +- 5% accuracy (N = number of readings taken).

Observed time (OT) = average of selected cycle times

Normal Time (NT) = OT x Rating factor / 100

Standard Time (ST) = NT + Allowances = NT x (1 + allowance fraction)

If allowances are x% of the TOTAL (standard) time: ST = NT / (1 - x)

* ALLOWANCES: personal needs (5-7%), basic fatigue (4%), variable fatigue (conditions of work), special / process allowance, contingency, delay allowance, policy allowance. Total typically 15 - 20%.

Rating techniques

* RATING = assessing the speed (tempo) of the worker relative to the NORMAL (standard) pace which is defined as 100 (60 on the British 0-100 scale; walking 4 miles per hour on level ground, dealing 52 cards in 0.5 minute).

Method	Idea
Speed rating	Rate the pace alone against normal 100 (e.g. 120 for a fast worker)
Westinghouse (levelling) system	Four factors: skill, effort, conditions, consistency, each graded (superskill +0.15 ... poor -0.22); R = 1 + sum of factors
Synthetic rating (Morrow)	Compare element times with PMTS (e.g. MTM): R = PMTS time / actual time
Objective rating (Mundel)	Pace rating + job difficulty adjustment factor
Physiological	Measures heart rate / oxygen consumption

Eg: Observed time of an element = 0.80 min, rating = 120, allowance = 15% (of normal time). NT = 0.80 x 1.20 = 0.96 min; ST = 0.96 x 1.15 = 1.104 min; standard output per 8-hour day = 480/1.104 = 435 pieces.
Eg 2: if the 15% allowance is on the total (standard) time: ST = 0.96 / (1 - 0.15) = 1.129 min.

Numerical example - time study

A job has 4 elements. Average observed times (min) and ratings are given. Allowances = 12% of normal time.

Element	Avg. observed (min)	Rating	Normal time
1	0.50	100	0.500
2	0.80	110	0.880
3	0.30	90	0.270
4	0.60	120	0.720
Total			2.370

Standard time = 2.370 x 1.12 = 2.654 min per piece

Output per 8 hour shift = 480 / 2.654 = 180.9, i.e. about 181 pieces.

Work sampling (ratio delay study)

* Statistical technique of making a LARGE number of RANDOM instantaneous observations of a worker / machine over a period to find the proportion of time spent in each activity (working, idle, delay). Developed by L.H.C. Tippett (UK, textile mills, 1934) - 'snap reading method'.
* Advantages: no stopwatch, many workers can be studied by one observer, less costly, no continuous watching, can be interrupted, good for non-repetitive jobs and idle time of machines. Disadvantages: gives no details of method, less suitable for short cycle jobs, observer error.

Sample size $N = 4\,p\,(1-p) / (L)^2$ (95% confidence, L = absolute accura

Eg: p = 0.30 (30% idle), required accuracy +-5% -> N = 4 (0.3)(0.7) / (0.05)^2 = 336 observations.

Standard time = (total time x % working x rating) / number of units produced x (1 + allowance).

Predetermined Motion Time Systems (PMTS)

* PMTS = techniques where STANDARD times (established from film studies) for basic human MOTIONS (reach, move, grasp, position, release, body motions) are used to build up the time for a job without stopwatch - the time is predetermined from tables of the motions involved. Times in TMU (time measurement unit): 1 TMU = 0.00001 hour = 0.0006 min = 0.036 second. Used even at the planning (design) stage, and no rating is needed.
* Systems: Work Factor (WF), Methods Time Measurement (MTM-1,

MTM-2, MTM-3), Basic Motion Time Study (BMT), Dimensional Motion Times (DMT), Master Standard Data, Modular Arrangement of PMTS (MODAPTS), Maynard operation sequence technique (MOST).

Methods Time Measurement (MTM)

* Developed by H. B. Maynard, G. J. Stegemerten and J. L. Schwab (Westinghouse, 1948). Each motion - REACH (R), MOVE (M), TURN (T) and APPLY PRESSURE, GRASP (G), POSITION (P), RELEASE (RL), DISENGAGE (D), eye travel and eye focus, body, leg and foot motions, crank (C), is analysed by DISTANCE moved, CASE (type), and WEIGHT, and its time is read from a table in TMU. 1 TMU = 0.00001 hour.
* Eg: Reach 20 cm case B -> about 12 TMU; Move 20 cm with 2 kg -> 13 TMU. Total TMU x 0.0006 = minutes; standard time includes allowance.

Work Factor system

* Developed by Quick, Shea and Koehler (RCA, 1938). Time for a motion depends on body member used (finger, hand, forearm, arm, trunk, leg, foot), DISTANCE moved, and WORK FACTORS - extra time for difficulty: (1) manual control - steering (S), directional control (U), precaution (P), definite stop (D), change of direction (C); (2) weight / resistance. Time in work-factor units; 1 WF unit = 0.0001 min (detailed) in Detailed WF; Ready WF, Abbreviated WF, Brief WF versions.

MTM	Work Factor
Motion analysed by distance, case, weight	By body member, distance, number of work factors
Unit TMU = 0.0006 min	Unit = 0.0001 min (detailed)
Rating not needed (tables = normal)	Rating inherent in the tables
More widely used	Developed by RCA; used for light assembly

4. Ergonomics

* ERGONOMICS (Human factors engineering; Greek ergon = work, nomos = law) = science of fitting the JOB to the WORKER - the study of the interaction between man, machine and working environment to improve comfort, efficiency, health and safety.

Principles of ergonomics

* 1. Match the workstation, tools and controls to the body dimensions (ANTHROPOMETRY - 5th to 95th percentile of the population). 2. Keep the posture neutral; avoid awkward, static, twisted or repetitive postures. 3. Reduce excessive force; use mechanical aids. 4. Keep everything in easy REACH - normal and maximum work areas. 5. Provide adequate clearance, adjustable seats and benches.
* 6. Control the ENVIRONMENT: lighting (glare free, 300 - 500 lux for assembly), noise (< 85 dB), temperature and humidity, ventilation, vibration. 7. Display design: clear, readable dials, colour coding, compatibility between control movement and display response. 8. Minimise fatigue: work-rest schedule, job rotation, task variety. 9. Safety: guards, emergency stops, fail-safe design. 10. Involve users:

participatory ergonomics.

* Areas: workstation design (sitting / standing), man-machine system, display and controls, lifting and manual handling (NIOSH lifting equation), cognitive ergonomics, office ergonomics (VDU, chair, keyboard). Benefits: higher productivity and quality, fewer accidents, less absenteeism and musculoskeletal disorders (back pain, carpal tunnel syndrome).

Anthropometry and workplace design

Item	Recommended
Normal work area (arm sweep, elbow fixed)	About 25 - 40 cm radius (hand); maximum area 60 - 70 cm
Sitting work surface height	About 70 - 75 cm (elbow height) for light work
Standing work height	Heavy work 10 - 15 cm below elbow; precision work 5 - 10 cm above
Seat	Adjustable height, lumbar (back) support, foot rest
Illumination (lux)	General 150, assembly 300 - 500, fine inspection 1000+
Noise	Below 85 dB(A) for 8 h exposure

Comparison of method study and work measurement

Method study	Work measurement
Finds the BEST METHOD of doing the job	Finds the TIME for the best method
Done first	Done after the method is standardised
Tools: charts, diagrams, SREDIM	Tools: stopwatch, work sampling, PMTS
Aim: eliminate waste motion	Aim: set time standard

Important questions

* 1. Define work study; explain method study procedure with SREDIM.
* 2. Explain flow process chart symbols; differentiate flow diagram and string diagram.
* 3. What are therbligs? Explain principles of motion economy.
* 4. Explain time study procedure; rating techniques; compute standard time.
* 5. Explain work sampling and its merits; PMTS, MTM and work factor system.
* 6. Explain principles of ergonomics.

Revision: Work study = method study + work measurement. SREDIM. ST = NT + allowance, NT = OT x rating. Work sampling - random observations. PMTS (MTM, WF) - TMU. Therblig - 17 basic motions.